pandas dataframe manipulation cheatsheet

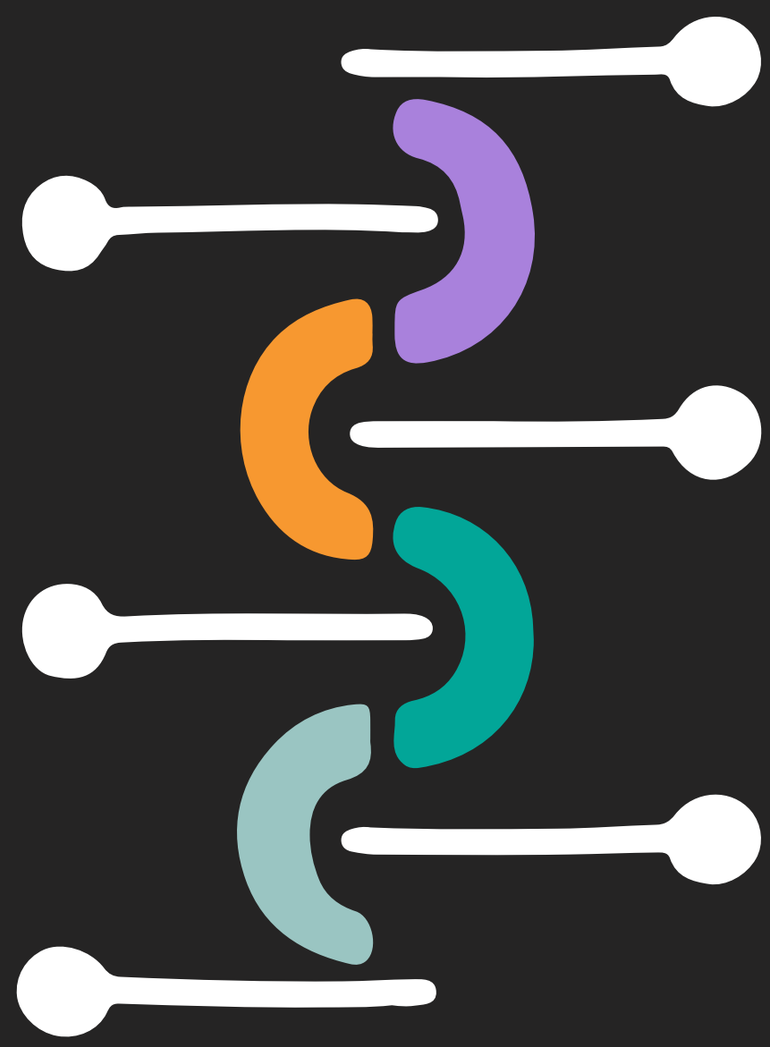
stratascratch.com

data exploring

tail()
helps you to show the last 'n' rows of the DataFrame

head()
helps you to show the first 'n' rows of the DataFrame

sample()
generates 'n' sample of your DataFrame



shape()
shows the dimension of the DataFrame

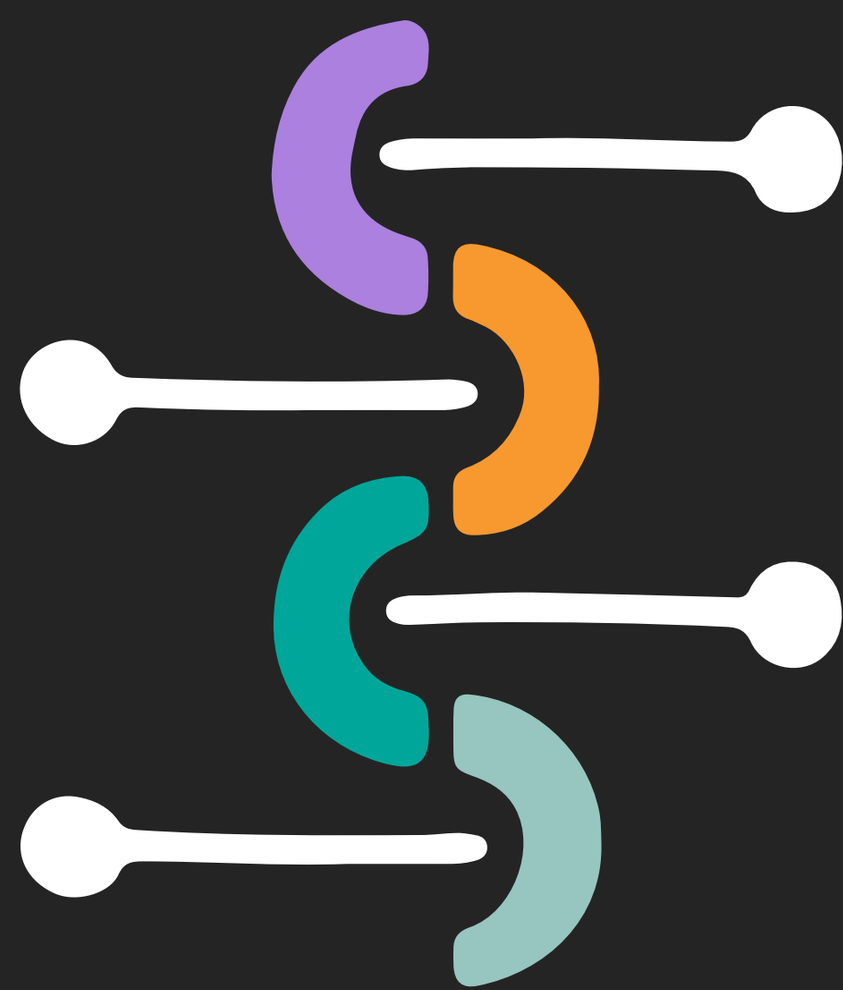
info()
gives you the column length, types and count of the column information of the Dataframe

describe()
gives descriptive statistics of your DataFrame: min,max, iqr,mean.

data retrieving

groupby()
it groups the DataFrame accordingly to the arguments

iloc()
it gets values of specific indexes



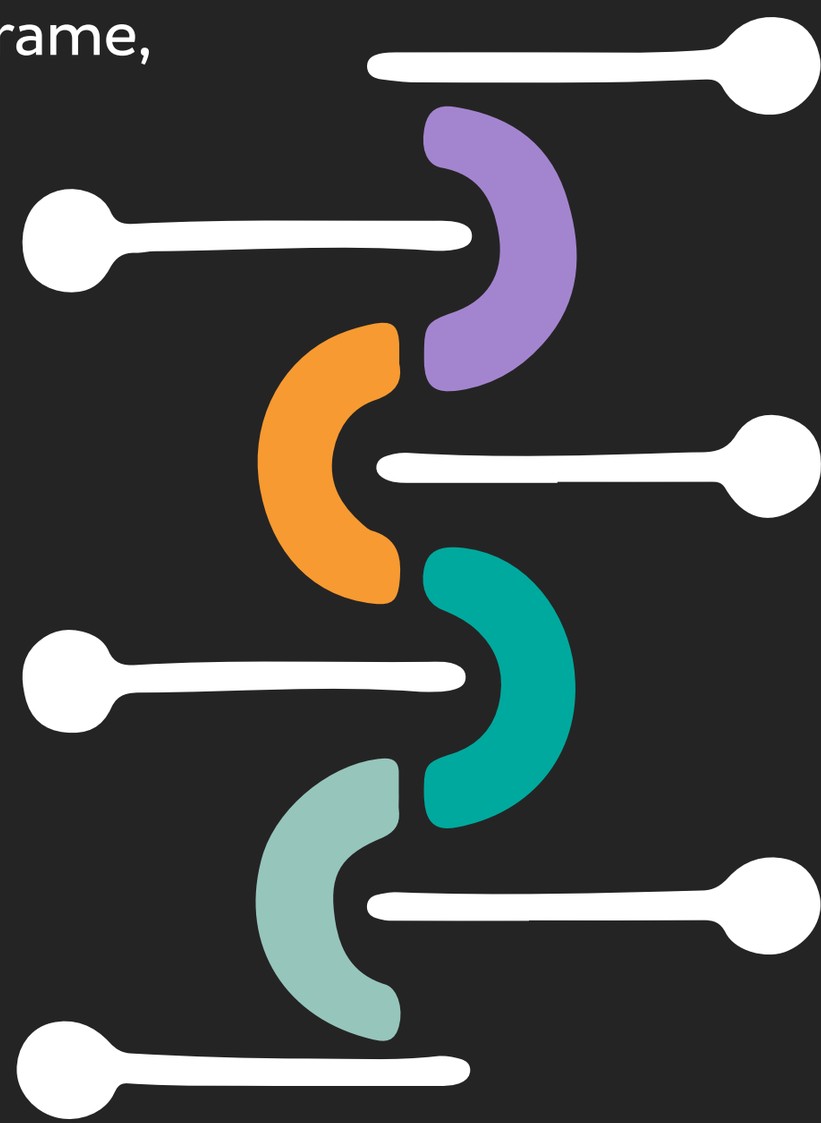
loc()
it helps you select the row and columns by names and labels

sort_values()
it sorts the DataFrame ascendingly or descendingly by a given comun

data operations

agg()
passes one or multiple functions to column/rows and renames the index of the resulting DataFrame, which can reproduce aggregare results

min()
calculates the min



max()
calculates the max

drop()
removes rows or columns by labeling name or location (index)

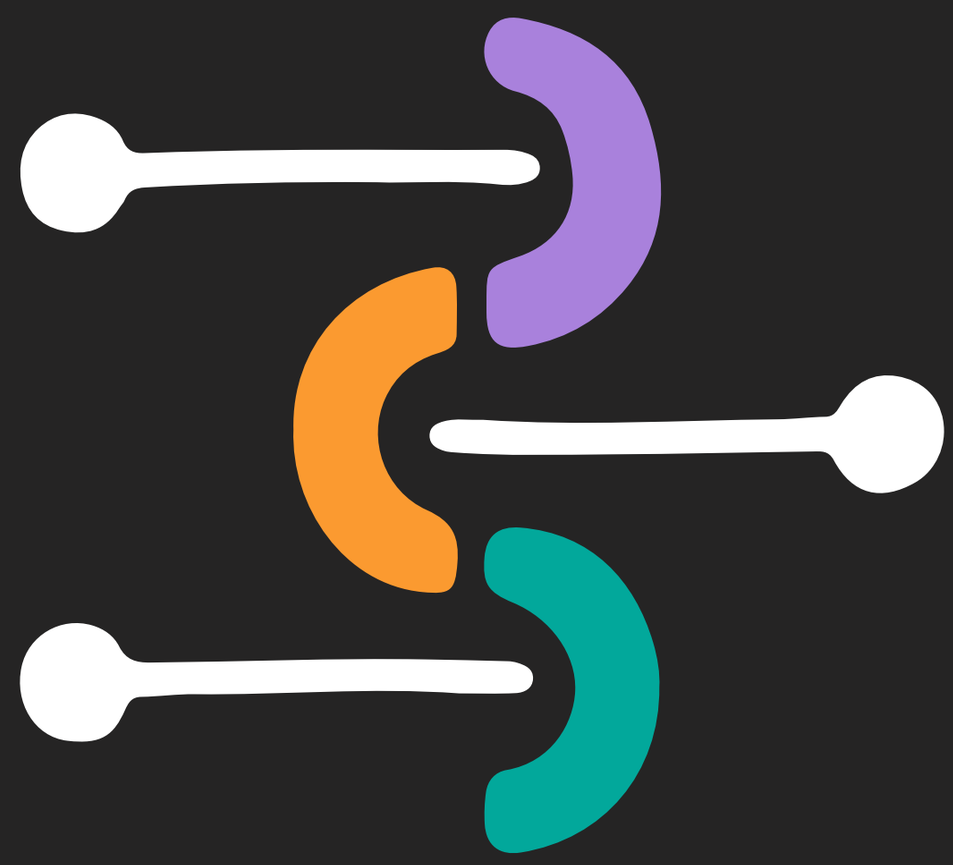
transform()
allows you to broadcast function and functions to column/rows, can not produce aggregated results, works with function or function list written in a list, dictionaruy or string like

apply()
performs operations on columns, rows/DataFrames, only allowed to work with functions & can produce aggregate results

data operations

drop_duplicates()
output DataFrame without duplicate rows

not_null()
detects non-missing values and returns booleans to show that

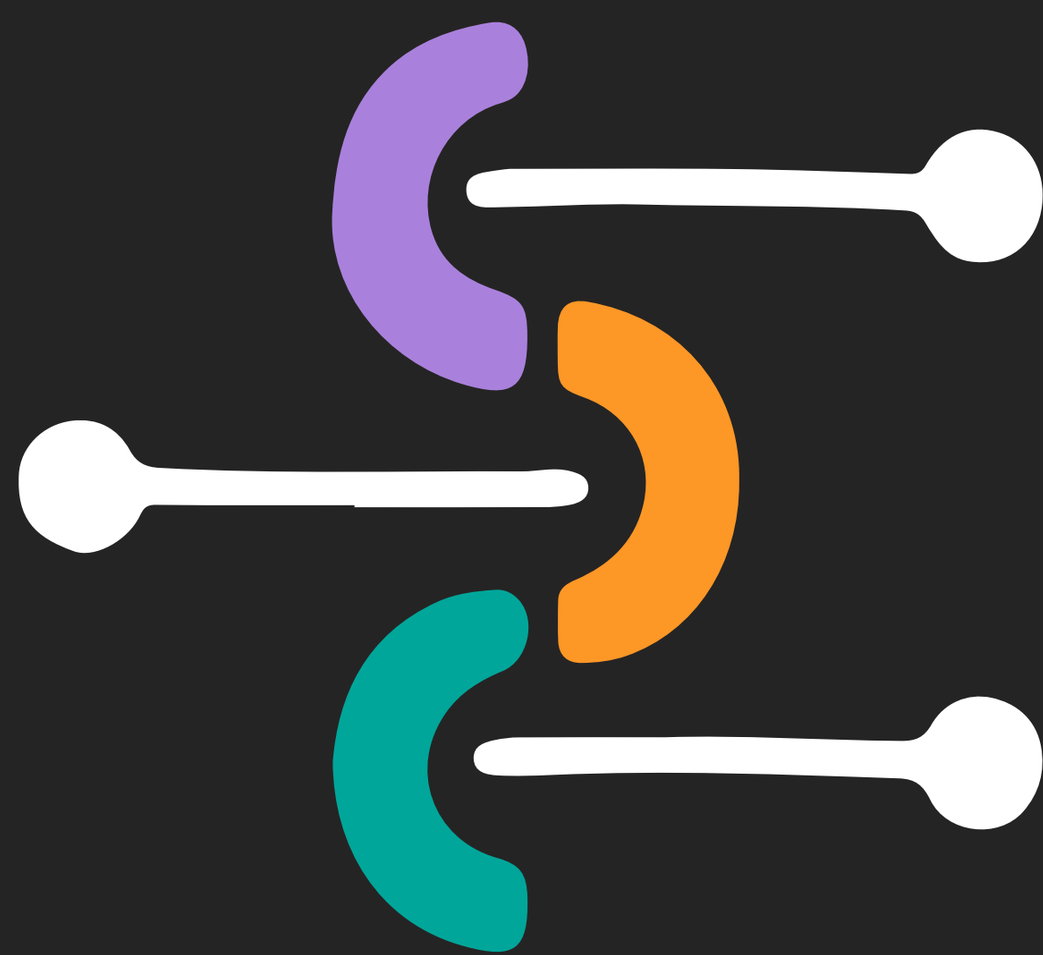


mean()
calculates the mean

missing value & NA's

isna()
returns booleans(true,false) which indicates the value na or not

dropna()
drops missing values from rows or columns

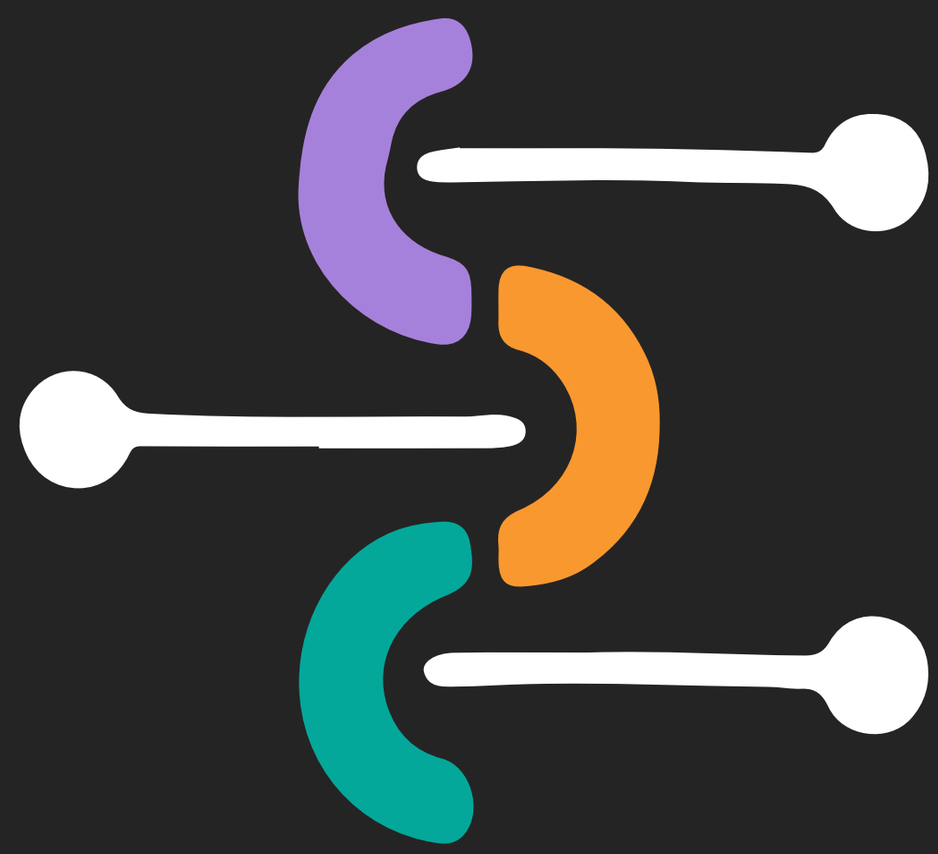


fill(na)
assigns specific values to NA's

merging

merge()
combines two DataFrames
on columns or indices

concat()
bonds two DataFrame
across the rows or columns

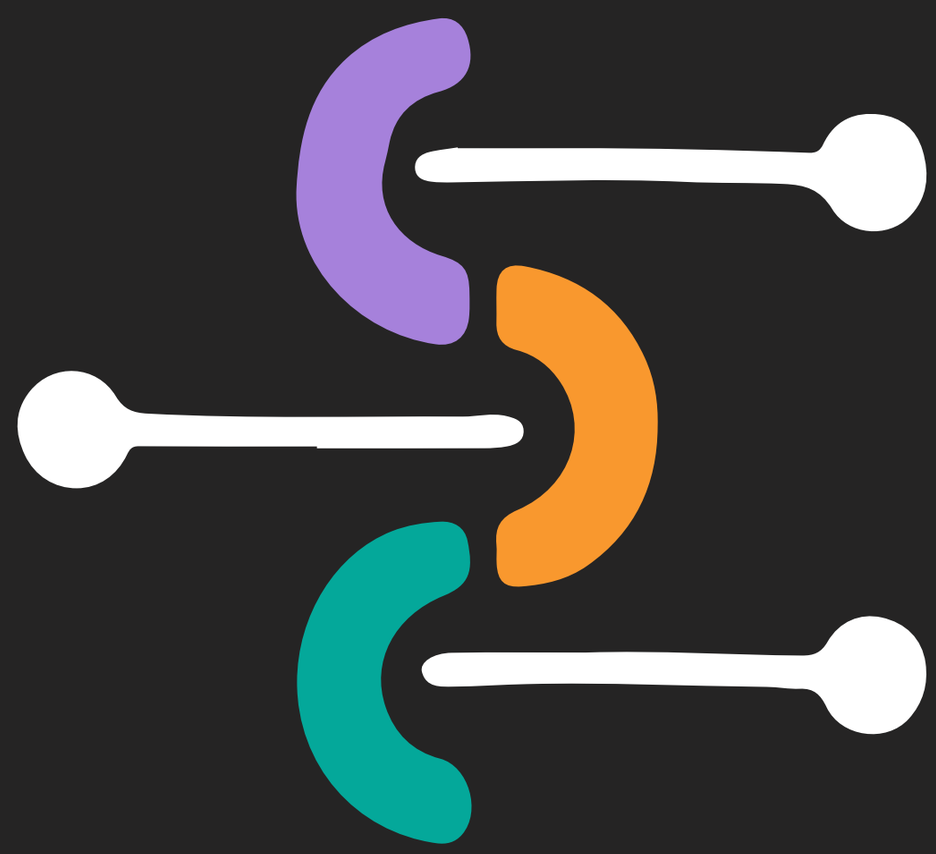


join()
joins two DataFrames
on a key column or index

plotting

plot.bar()
to plot bar graph

plot.line()
to plot line graph



plot.scatter()
to draw scatter plot

plot.pie()
top lot pie chart

saving from and reading to DataFrame

read_sql() To read from SQL into a DataFrame	**to_sql()** To save the DataFrame into a SQL
read_excel() To read from excel files into a DataFrame	**to_excel()** To save the DataFrame into a CSV file
read_csv() To read from CSV files into a DataFrame	**to_csv()** To save the DataFrame into a CSV file
read_json() To read from json files into a DataFrame	**read_json()** To save the DataFrame into a json file

date time

dt.month()

This function returns the number of month

dt.quarter()

It selects the “n” quarter of the DataFrame.
2 means the first 6 months

to_datetime()

It changes the format of the DataFrame/series to a pandas date time object.

dt.year()

It returns the year of the date.